Background on Stochastic Differential Equations, Density Evolution, and Score Based Diffusion

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Overview

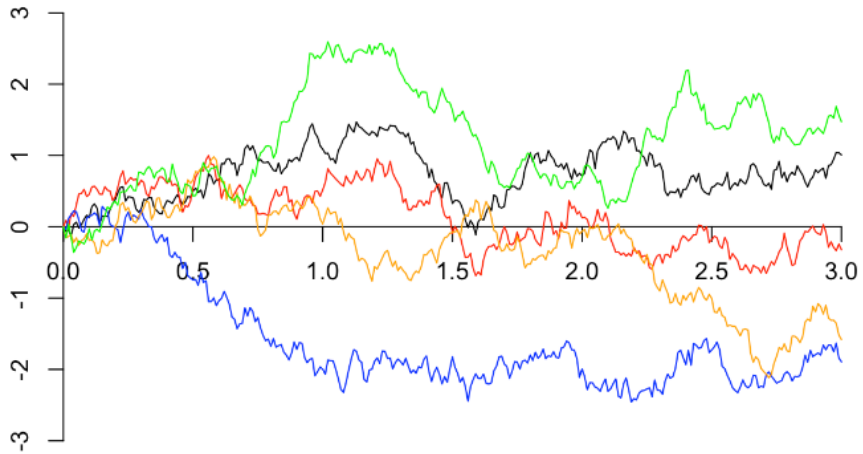
- Introduce relevant topics in continuous time stochastic processes
- Stochastic Differential Equations (SDE)
- Kolmogorov Fokker Planck Equation of Density Evolution
- A Case Study on the Langevin Diffusion
- Forward and Backward SDE's of Score Based Diffusion

Stochastic Differential Equations

Brownian Motion

- A continuous time stochastic process $\{X_t\}_{t\geq 0}$ is a sequence of random variables X_t indexed by a continuous time index $t \in \mathbb{R}_{\geq 0}$. It is defined by it's joint measures $P(X_{t_1}, \cdots, X_{t_k})$ at any sequence of finite times.
- The basic stochastic process is Brownian Motion $\{B_t\}_{t\geq 0}$ which is the continuous time stochastic process with the following properties:

1. $B_0 = 0$
2. Continuous sample paths, $t \to B_t$ is continuous
3. Independent increments $(B_{t_1}, B_{t_2} - B_{t_1}, \cdots, B_{t_k} - B_{t_{k-1}})$
4. $B_{t+s}|B_t \sim N(B_t, s)$



More General Processes

- In our study of stochastic processes, we are most interested in the marginal distributions at time t, $X_t \sim p_t$.
- The stochastic processes we want to study are defined by so called "*stochastic differential equations*"

$$dX_t = V(X_t, t)dt + G(X_t, t)dB_t, \quad X_0 \sim p_0$$

- We now go about making sense what process an SDE is describing.
- Our goal is to reach a "functional/ working understanding" of how to work with these SDE rather than the technical details of the measure theory "behind the scenes."
- Focus on how marginal distributions $X_t \sim p_t$ evolve with time for an SDE.

(Some) Measure Theory

- In a measure theory sense, we have a underlying probability space Ω, a sigma algebra $\mathcal{F}$, and underlying measure μ.
- A "random variable" is a measurable function of the underlying probability space to the state space, in our case $\mathbb{R}^d$.
- Each of these B_t as a random variable is simply a mapping from $w \in \Omega \to \mathbb{R}^d$. We think of $B_t(\omega)$.
- There is also an increasing family of sigma algebras (filtration) $\{\mathcal{F}_t\}_{t \geq 0}$ under which B_t is $\mathcal{F}_t$ measurable.
- So we can think of the whole stochastic process as defined by the mapping $B : \mathbb{R}_{\geq 0} \times \Omega \to \mathbb{R}^d$, $(t, \omega) \to B_t(\omega)$.
- For a fixed ω, $t \mapsto B_t(\omega)$ is one realized sample path of the process, for Brownian motion these functions are continuous.

Stochastic Integrals

- The definition of an SDE follows first from the idea of a "stochastic integral."
- Say we have two stochastic processes $\{G_t\}_{t\geq 0}$ and $\{W_t\}_{t\geq 0}$ which are called "progressively measurable" (G_t "only depends on events in time [0,t]" or is measurable with respect to the sigma field defined up to time t).
- We define the stochastic integral as a new continuous time stochastic process denoted as

$$S_T \equiv \int_0^T G_t dW_t$$

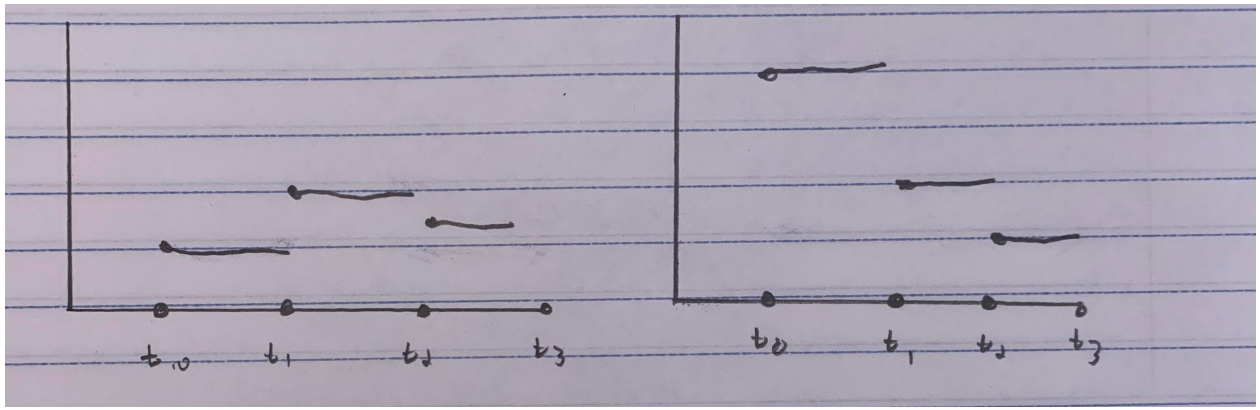
- This means that $\{S_t\}_{t\geq 0}$ is a sequence of random variables, not a deterministic number. It is itself a mapping $S : \mathbb{R}_{\geq 0} \times \omega :\to \mathbb{R}^d, (t, \omega) \to S_t(\omega)$.

Step Process

- We first define the stochastic integral for step processes. Consider if we have k independent random variables H_i and time increments $0 = t_0 < t_1 < \cdots < t_k = T$. Define the "step" stochastic process

$$G_t(\omega) = \sum_{i=1}^{K} H_i(\omega) 1\{t_{i-1} \leq t < t_i\}$$

- Essentially, we pick levels H_i randomly and then the sample paths are a step process between these levels.



Stochastic Integral of Step Functions

- For a step process, we define the stochastic integral as the random variable

$$\int_0^T G_t dW_t = \sum_{i=0}^{K-1} H_i (W_{t_{i+1}} - W_{t_i})$$

- That is, it is like a "random area" where the heights of the rectangles are determined by the levels of G_t, while the "width" of the area is determined by W_t.
- For a step process, this is a fully well defined stochastic process.
- Covering up a lot of details here, but for a general stochastic process, G_t, consider it as the limit of a series of step processes $G_t = \lim_{l \to \infty} G_t^{(l)}$ and define the stochatic integral as the limit of integrals of step functions

$$\int_0^T G_t dW_t = \lim_{l \to \infty} \int G_t^{(l)} dW_t$$

SDE From Stochastic Integral

- When we state an SDE, we define it in terms of a stochastic integral. The stochastic process $(X_t)_{t\geq 0}$ which follows the SDE

$$dX_t = v(X_t, t)dt + G(X_t, t)dW_t, \quad X_0 \sim p_0$$

- Where $v : \mathbb{R}^d \times \mathbb{R}_{\geq 0} \to \mathbb{R}^d$ is a vector valued function called the drift and $G : \mathbb{R}^d \times \mathbb{R}_{\geq 0} \to \mathbb{R}^{d\times d}$ returns a matrix which defines a covariance called the dispersion $\sigma(X_t, t) = \frac{1}{2}G(X_t, t)G(X_t, t)^T$.
- The process is **defined as** the stochastic process which satisfies the equation

$$X_T = X_0 + \int_0^T v(X_t, t)dt + \int_0^T G(X_t, t)dW_t$$

- In a discrete time approximation, we think of

$$X_{t+\delta} = X_t + \delta v(X_t, t) + \sqrt{\delta}G(X_t, t)Z, \quad Z \sim N(0, I)$$

Kolmogorov Fokker Plank Equation

- Say we have a SDE defined by $dX_t = v(X_t, t)dt + G(X_t, t)dB_t$ with B_t being Brownian motion. We say X_t has marginal density $p_t(x)$. Then the time evolution of this density is defined by the Kolmogorov Fokker Planck equation

$$\frac{d}{dt}p_t(x)$$
$$= -\sum_{i=1}^{d} \frac{\partial}{\partial x_i}[p_t(x)v_i(x,t)] + \frac{1}{2}\sum_{i=1,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j}[p_t(x)(G(x,t)G(x,t)^T)_{i,j}]$$
$$\equiv -\nabla^T(p_t(x)v(x,t)) + \langle \nabla^2, p_t(x)\frac{1}{2}G(x,t)G(x,t)^T \rangle_{HS}$$

- This is the negative divergence (sum of partial derivatives) of the density times the drift plus sum of the second derivatives of the density times the dispersion.
- Note that if $G(X_t, t)$ is diagonal, then there are no cross terms and the second term above is the Laplacian (trace of hessian).

Proof of KFP

(see attached)
To simplify things, we usually consider constant dispersion

$$dX_t = v(X_t, t)dt + dB_t$$

in which case the KFP is the negative divergence of drift times density plus half the Laplacian of the density

$$\frac{d}{dt}p_t(x) = -\nabla^T[p_t(x)v(x,t)] + \frac{1}{2}\Delta(p_t(x))$$

Invariant Measure of SDE

- Consider now an *SDE* where the drift is not time varying,

$$dX_t = v(X_t)dt + dB_t$$

- The invariant measure of an SDE is a density $p^*(x)$ such that if $X_t \sim p^*(x)$, then $X_T \sim p^*(x)$ for all $T \geq t$.
- This is the same as saying $\frac{d}{dt}p_t(x) = 0$ when $p_t(x) = p^*(x)$ for all x. That is the density is not changing anywhere.
- By KFP equation, the invariant measure of a given SDE is the function $p^*(x)$ which satisfies

$$0 = -\nabla^T(p^*(x)v(x)) + \frac{1}{2}\Delta p^*(x)$$

- Once we know the drift, we can determine the invariant measure of an SDE.

SDE For Sampling

- Say we have a density p^* we want to sample from. We can design an SDE which admits p^* as it's invariant measure via solving KFP equation equal to 0.
- Then if we run the SDE long enough, draw $X_0 \sim \mu$ from some prior and by time T samples X_T will be near samples from p^* (convergence rates matter here.)
- Idea though is we can design SDE's as a sampling process converging to their invariant measure.

Case Study: Langevin Diffusion on a Normal Target

Langevin Diffusion

- Say we have a density proportional to $p^*(x) \propto e^{f(x)}$ we want to sample from. The **Langevin Diffusion** is defined as the the SDE

$$dX_t = \nabla f(X_t)dt + \sqrt{2}dB_t$$

(drift term is **score**, the gradient of the log likelihood we want to sample from)

- Notice that we don't need normalizing constant of density to determine this diffusion, works for unnormalized targets.
- Invariant measure of diffusion is $e^{f(x)}$. Check KFP,

$$\begin{aligned}
-\nabla^T(p(x)\nabla f(X)) + \frac{1}{2}\Delta((2)p(x)) &= -\sum_{i=1}^{d}\frac{\partial}{\partial x_i}(e^{f(x)}\frac{\partial}{\partial x_i}f(x)) + \frac{1}{2}(2)\sum_{i=1}^{d}\frac{\partial^2}{\partial x_i^2}e^{f(x)} \\
&= -\sum_{i=1}^{d}\frac{\partial}{\partial x_i}(\frac{\partial}{\partial x_i}e^{f(x)}) + \sum_{i=1}^{d}\frac{\partial^2}{\partial x_i^2}e^{f(x)} \\
&= 0
\end{aligned}$$

Standard Normal Target

- Say our target is standard normal, $p^*(x) \propto e^{-\frac{1}{2}x^T x}$, then drift is just linear term and dispersion is constant

$$dX_t = -X_t dt + \sqrt{2} dB_t$$

- Also called the Ornstein Uhlenbeck Process.
- We can show that conditional distributions of this SDE are of the form

$$p(X_t|X_0) = N(e^{-t}X_0, (1 - e^{-2t})I_{d\times d})$$

- Note the rapid decay in dependence on X_0, so for any initial distribution $X_0 \sim \mu$ the marginal at time t will rapidly converge to standard normal

$$p_t(x) = \int p_t(x|x_0)\mu(x_0)dx_0 \to N(0, I)$$

Score Based Diffusion: Forward and Backward SDE

Backward Flow and Forward Flow Equations

- A remarkable result result by Anderson 1982 [1] shows that any SDE (which we might call the "forward flow" SDE) has a corresponding partner (the "backward flow" SDE) which has the exact opposite marginal density evolution.
- That is, with $X_0 \sim p_0$ and following the SDE

$$dX_t = f(X_t, t)dt + g(t)dB_t$$

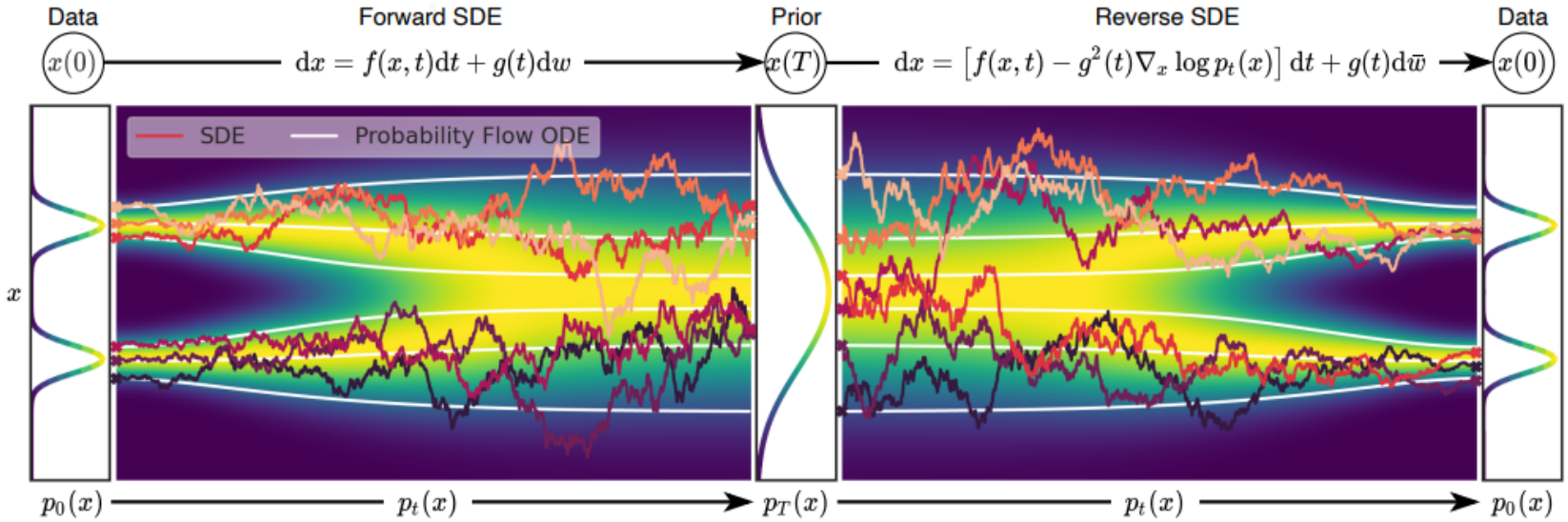
say each X_t has marginal distribution p_t. So this process evolves the marginals from $X_0 \sim p_0$ to $X_T \sim p_T$

- There is a sister equation with $Y_0 \sim q_0 = p_T$ and

$$dY_t = -[f(Y_t, T-t) - g(T-t)^2 \nabla \log p_{T-t}(Y_t)]dt + g(T-t)dB_t$$

which has marginals $Y_t \sim q_t = p_{T-t}$. That is, the forward flow evolves from marginal p_0 to p_T while the backward flow evolves from p_T to p_0.

Score Based Diffusion



[2]

- The drift of the backward SDE is the negative difference of the drift of the forward flow SDE and the **score** of the marginal density at time $T - t$.

$$dY_t = -[f(Y_t, T-t) - g(t)^2 \nabla \log p_{T-t}(Y_t)]dt + g(t)dB_t$$

- This gives rise to the name score based diffusions, a "backward flow" SDE defined by the score of a forward flow SDE.

Backward Flow Proof

(See document)

Score Based Diffusion as a Sampling Method

- Say we have some complex density p^* we want to sample from. In theory, it is "easy" to evolve from this complex density to a simple density, just run Ornstein Uhlenbeck and we will quickly get samples from pure noise (standard normal)

$$dX_t = -X_t dt + \sqrt{2} dB_t, X_0 \sim p^*$$

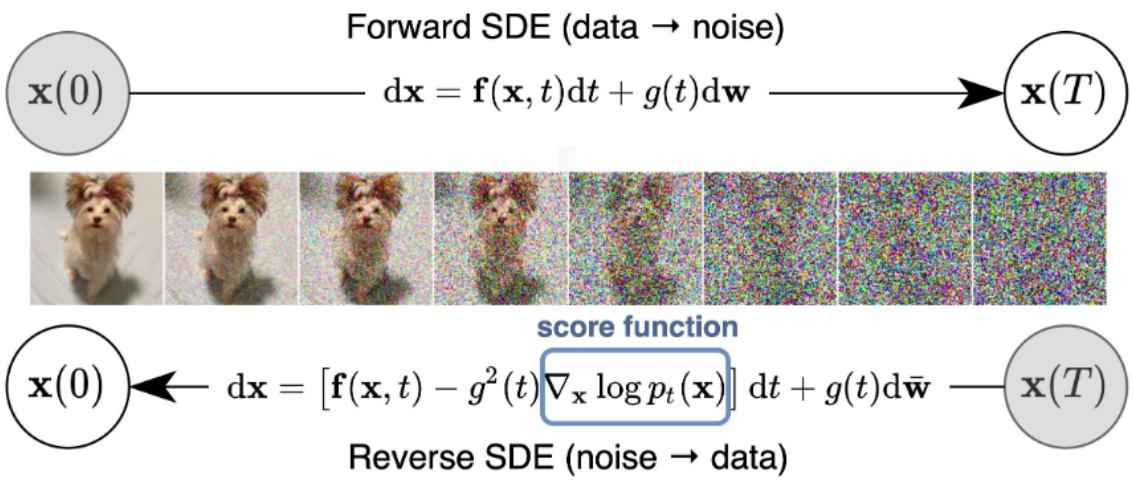
  with $X_T \sim N(0, I)$ for some large T.
- Yet this then defines a process which would take us from a standard normal sample back to our complex target. Thus, if we can compute the scores $\nabla \log p_t(X)$ under our forward flow, we can run the backward flow with $Y_0 \sim N(0, I)$ and

$$dY_t = -[f(Y_t, T - t) - 2\nabla \log p_{T-t}(Y_t)]dt + \sqrt{2} dB_t, Y_0 \sim N(0, I)$$

  and terminate with $Y_T \sim p^*$.
- We don't actually implement the forward flow, it is just a conceptual process which defines the scores which we actually implement with the reverse flow

Example Application

- Let's say you want to generate images of dogs in 1024x1024 pixel images. If each pixel can take values $x_i \in [-1, 1]^4$ RBGY, every image is defined by $1024x1024x4$ numbers. Consider $p_{\text{dog}}(x)$ as a distribution over all images describing what an image of a dog looks like.
- The forward flow process amounts to "adding noise" to our image to make it look more like random noise
- The reverse flow process takes in random noise and converts it into a new image of a dog, sampled from the distribution of dog images but not any specific image of a dog seen before. [2]



References


B. D. Anderson, "Reverse-time diffusion equation models," *Stochastic Processes and their Applications*, vol. 12, no. 3, pp. 313–326, 1982.


Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," *arXiv preprint arXiv:2011.13456*, 2020.

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